

Problem

Find the complex Fourier series for $f(t) = e^{-t}$, $0 < t < 1$, with $f(t + n) = f(t)$ for all integer values of n .

Step-by-step solution

Step 1 of 1

$$\begin{aligned}
 T &= 1, \quad \omega_o = \frac{2\pi}{T} = 2\pi \\
 f(t) &= \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_o t} \\
 C_n &= \frac{1}{T} \int_0^T f(t) e^{-jn\omega_o t} dt \\
 C_n &= \int_0^1 e^{-t} e^{-2jn\pi t} dt = \int_0^1 e^{-(1+2jn\pi)t} dt \\
 &= \left. \frac{e^{-(1+2jn\pi)t}}{-(1+2jn\pi)} \right|_0^1 \\
 &= \frac{e^{-(1+2jn\pi)}}{-(1+2jn\pi)} - \frac{1}{-1+2jn\pi} \\
 &= \frac{e e^{-2jn\pi}}{1+2jn\pi} (e-1) \quad : \boxed{e^{-2jn\pi} = \cos 2n\pi - j \sin 2n\pi = 1} \\
 f(t) &= (e-1) \sum_{n=-\infty}^{\infty} \frac{e^{+j2n\pi t}}{(1+2jn\pi)}
 \end{aligned}$$